

Lesson 34: Conics in Polar Coordinates

Criteria	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Identification & Classification	Accurately identifies the conic type (ellipse, parabola, hyperbola) and eccentricity (e) from any polar form.	Identifies conic type and e from standard polar form ($1 \pm e \cdot \cos(\theta)$).	Identifies conic type, but makes minor errors in calculating e.	Cannot identify the type of conic from the polar equation.
Equation Conversion	Successfully converts between polar equations $r = \frac{ed}{1 \pm e \cos \theta}$ and standard rectangular equations (x, y).	Converts between forms with minor algebraic errors.	Understands conversion process but struggles with algebraic manipulation.	Incorrectly identifies standard form components.
Directrix & Orientation	Identifies type (vertical/horizontal) and position (d) of the directrix, including sign in denominator.	Correctly identifies orientation and d value correctly.	Identifies orientation but mixes up vertical/horizontal or signs.	Incorrectly identifies directrix line.
Graphing & Key Points	Plots accurate graphs on polar paper, correctly plotting vertices and key points based on polar coordinates.	Graphs with correct orientation and shape, with minor plotting inaccuracies.	Sketch is rough; key vertices are missing or improperly plotted.	Plotting is incorrect or does not represent the conic.

Analysis (Foci/Vertices)	Accurately identifies key features: focus at origin, vertices, and center (if applicable).	Identifies key features with minor errors.	Identifies focus, but struggles to find vertices in polar form.	Unable to identify key features of the conic.
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